

Process Control: Level Control

Pre-Lab Plan

Unit Operations Lab 6

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Group #2, Tuesday, PM section

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1. Experimental Objective (5pts)

The objective of this experiment is to demonstrate and analyze the use of process control system Armfield PCT50 Level Control by using a single PID loop to measure the level of the water fluid as the measurable variable. Disturbances will be introduced in the process, which will require to change the pump speed or the valve positioning, resulting in changes of the water level in the tank. The goal is to use the control elements (pump speed and valve positioning) to maintain a steady water level in the reservoir.

All the data needed for further analysis will be recorded by the software PCT50 Software. For this experiment, the independent variables will be the pump speed, the orifice size on solenoid, control valve 2, while the dependent variables are the level of water and the flowrate.

2. Experimental

The apparatus and the materials and supply of this experiment are introduced below and briefly explained their usage.

2.1 Apparatus (5 pts)



Figure 1: Picture of the Apparatus used in the Process Control Experiment.

Table 1: Apparatus Instrument and Briefly Description

Number	Instrument	Description
1	Tank	Reservoir where the level of water is been controlled
2	Sump Pump	Equipment used to pump the water to the top tank
3	PID-Controller	Proportional Integral Derivative Controller
4	CV1	Control valve 1, which allows water to be pumped out of the system
5	CV2	Control valve 2, which allows water into the system
6	CV3	Control vale 3, which allows water into the system
7	Solenoid	Device that controls the water flowing through CV3
8	Computer	Software connected to the controller
9	Drainage Hose	Hose used to drain the water in the system after procedure.

2.2 Materials and Supplies (5 pts)

Table 2: Materials & Supplies

Number	Material	Description
1	Water	Fluid used to test the PCT50 level control
2	Graduated Cylinder	Device used to transport water to the apparatus

2.3 Experimental Plan (10 pts)

- **Setup**

1. Make sure the drain valve located at the rear of the sump tank is completely closed
2. By pouring clean water through the cut out at the front, fill the sump tank until water is 30 mm below the cut out region in normal use.
3. Disconnect the flexible tube from the pump outlet, let the pump run for a short period of time at high speed. Reconnect the tube.
4. Open the valve CV2 outlet completely and fit a 4mm diameter orifice below it, followed by a 2mm orifice below the solenoid valve.
5. Connect the USB socket at the back of the electrical enclosure of the computer.
6. Run the PCT50 software and select Experiment Ex1: Inflow Control.
7. Use the red telephone icon in the top toolbar to make sure the Virtual COM port has been enabled.
8. Adjust inlet valve CV1 in order to obtain the proper flow into the process reservoir by fully opening both the outlet valve CV2 and the inlet valve CV1.
9. Choose the "View \ Diagram" icon to display a mimic diagram of the process.
10. Click Power on.
11. If the reading is not 0mm for an empty vessel, zero the reading by pushing the zero button.
12. Set the pump speed to 50%.
13. Adjust CV1 until the water level remains steady around the mid height in the process reservoir. This means that the inlet flowrate at CV1 has matched the outlet flowrate at CV2.
14. Reset the pump speed to 0% on the mimic diagram. Let the vessel drain by allowing fluid through outlet CV2.

- **Experiment A : Manual Control of Water Level (Open Loop)**

1. With a constant level in the reservoir, choose the "GO" icon to start recording the data, note the level reading L1 in mm.
2. By clicking in the mimic diagram, open the solenoid valve. "1" indicates that the valve is open.
3. Notice the decrease in the level, due to an additional outflow via the solenoid valve.
4. When the level indicates a fall of 20mm, note the current pump speed on the mimic diagram.
5. Increase the pump speed in order to restore the original level.
6. Close the solenoid valve by clicking the button on the mimic diagram. "0" indicates that the valve is closed.
7. Observe the level increasing. When the level increases by 20mm, note the pump speed on the mimic diagram.
8. Reduce the pump speed to restore the original level.
9. Click the "STOP" icon to end the collection of data.
10. Click the "View \ Graph" to display a graph with the data recorded.
11. Observe the water level as the solenoid was opened and closed, and the action needed in order to restore the level in the reservoir.
12. Return to the mimic diagram, allow the process reservoir to drain by reducing the pump speed.
13. Create a new result sheet by clicking in the "Table" icon in the tool bar.
14. Choose the "GO" icon to start recording data.
15. Gradually increase the speed of the pump until the level start to rise.

16. At 150mm mark, stop the pump and allow the reservoir to drain.
17. After the reservoir empty it out, set the pump to 75% speed and allow the vessel to fill up.
18. At the 150mm mark, stop the pump and let the reservoir to drain.
19. Click the "STOP" icon to stop recording the data.
20. View and observe the graph at the different rates of filling vs different pump speeds. Note that the vessel drains at the same rate.
21. If time allows, repeat this procedure for different sized of orifices. Ex: 2mm at CV2 and 4mm at the solenoid valve. Different settings of CV1 can also be investigated.
22. Create a new results sheet for each new run, by selecting the "Table" icon in the tool bar.
23. Save a copy of the results and graphs.

• **Experiment C : Proportional Control of Water Level (Closed Loos, P only)**

1. Make sure the equipment has been property set up by following the Se-up steps.
2. Click on the PID box in the mimic diagram. Set the Proportional Band P to 200%, the Integral Time I to 0 and the Derivation time D to 0. Set the Set Point to 75mm.
3. Click Apply to enter the modifications.
4. Click on the Automatic Mode of Operations.
5. Click on the GO icon to start collecting data.
6. The water level will start to rise after the pump starts to work. The steady level will not correspond with the setpoint of 75mm due to the offset inherent in a P only controller. So open the solenoid valve by clicking on the mimic diagram (1 indicates that the valve is open).
7. Observe the level decreasing and the pump speeding up.
8. Allow the level to settle, then close the solenoid valve and allow the level to resettle.
9. Adjust the setpoint to 100mm. Click apply.
10. Observe the response of the system.
11. After the water level has settled, return the setpoint to 75mm.
12. Observe the response.
13. Adjust the P term to 100% in the PID controller. Click apply.
14. Observe that the pump speed has increased, and the offset reduces.
15. Open then close the solenoid valve as done previously.
16. Observe the response of the system.
17. Adjust the P term to 50%, and repeat the previous steps.
18. Adjust the P term to 20% and repeat the previous steps.
19. Adjust the P term to 10% and repeat the previous steps.
20. Adjust the P term to 5% and repeat the previous steps.
21. Adjust the P term to 2% and repeat the previous steps.
22. Click on STOP to stop recording the data.
23. Go to View \ Graph to display the data obtained.
24. Click on Format \ Graph Data and Plot Level L1, Set point on 1st axis, and Controller Output on the 2nd axis.
25. Observe the changes in water level and the stability of the responses for each variation on P or change in set point.
26. Choose a setting for P, which gives a better stable control with a reasonably small offset for the next part of the experiment.

- **Experiment C : Proportional Control of Water Level (Closed Loos, P + I controller)**

1. Start up a new results sheet for this part of the experiment, by selecting the Table icon in the tool bar.
2. Choose the PID box on the mimic diagram, set the Proportional Band P to the value selected above, the Integral Time I to 0 seconds and the Derivation Time D to 0, followed by the Set Point to 75mm.
3. Click Apply.
4. Click the GO icon to start collecting data.
5. Choose the Automatic mode of operation and wait until the water level has stabilized.
6. Adjust the I term to 100 seconds in the PID controller. Click Apply.
7. Observe the pump speed.
8. After the water level has settled, open the solenoid valve to introduce disturbance.
9. Observe the water level decreasing, and the pump speed varying.
10. After the water level has resettled, close the solenoid and wait until the level settle once again.
11. Adjust the set point to 100mm. Click Apply.
12. Observe the response of the system.
13. After the water level has settle, return the set point to 75mm.
14. Observe the response of the system.
15. Adjust the I term to 50 seconds in the PID controller. Click Apply.
16. Observe the water level, and how long it takes to return to the set point.
17. After the level has settled, open then close the solenoid valve as previously explained.
18. Adjust the set point to 100mm. Click Apply.
19. Observe the response of the system.
20. After the level has settle, return the set point to 75mm.
21. Observe the response of the system.
22. Incrementally decrease the Integral time settings and repeat the disturbance and setpoint changes until the level becomes unstable.
23. Click on View \ Graph to display a graph of the data collected.
24. Choose Format \ Graph data and Plot Level L1 and Set point on 1st axis. Controller output on the 2nd axis.
25. Observe the changes in the water level, and how the system responds to changes in I setting with a disturbance or change in the set point.
26. Choose the setting for I, which has a better control and reduces any offset swiftly without allowing the system to be unstable.

Project Safety Evaluation (10 pts)

Project Title: Process Control: Level Control

Date: 11/12/2019

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Possibility of electrical shock due to electrical equipment and water.
Potential mechanical hazards?	Yes	If yes, identify conditions: Make sure to not run the pump dry.
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: Nitrile gloves
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water - ACC# 00199 (Fischer scientific)
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	25 °C	No temperature hazards, water is at room temperature.
Pressure (psia)	14.7 psia	Water runs at atmospheric pressure. No pressure hazards.
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	<i>yes</i>	<i>Shut off valves and power switches.</i>

I have read relevant background material for the Unit Operations Laboratory entitled: "Process Control: Level Control" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety

practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>James Fell</u>	11/12/2019
signature of team member	date

<u>Jasmine Villegas</u>	11/12/2019
signature of team member	date

<u>Mariana St. Peter</u>	11/12/2019
signature of team member	date

<u>William Rawson</u>	11/12/2019
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